

University of Illinois at Chicago

Department of Chemical Engineering

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SAFETY QUIZ: CHE381- Chemical Engineering Laboratory

Please answer the following questions testing your knowledge of the content of the Chemical Hygiene Plan safety document. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words.

1. What is the most fundamental rule of chemical safety?

Awareness.

2. What two pieces of personal protective equipment must be worn at all times in the lab?

Safety  googles and shoes.

3. What additional kinds of personal protective equipment are available depending on the chemical hazard or experimental conditions?

Lab coats, gloves, face shields, ear protection, splash aprons

4. True or False: Fume hoods are generally used as storage areas for chemicals.

false

5.  Fume hoods are to be used when the Threshold Limit Value (TLV) of a chemical is at or below:

True

6. What are the four routes of chemical entry into the body?\

inhalation, ingestion, injection, and eye or skin contact

7. True or False: Respirators are generally used in the R&D laboratories for normal work procedures.

False

8.  What is the role of the Chemical Hygiene Plan?

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The Chemical Hygiene Plan is designed as a tool to coordinate safety procedures. Every employee in the laboratory must be trained in the applicable details of this Plan and must comply with its provisions.

9. What is the guideline for laboratory room airflow and fume hood airflow?

General laboratory room ventilation should normally be not less than 20 linear feet per minute airflow, thus providing comfortable breathing air for workers.

Fume hoods should provide 80-120 linear feet per minute airflow

10. List 3 Common Housekeeping Practices that promote chemical hygiene and safety:

- Keep all aisles, hallways, and stairs clear of all chemicals.
- Keep all work areas and especially workbenches clear of clutter and obstructions.
- All working surfaces and floor should be cleaned regularly.

11. What is the proper method of transporting gas cylinders in the laboratory?

- Gas cylinders must be transported using 1-cylinder, 4-wheel carts.
- The cylinder cap must be on the gas cylinder during transportation

12. True or False: MSDS sheets are required only for toxic chemicals and gases used in the laboratory.

False

13. What is an accepted method of vapor detection in the laboratory?

If suspicious odors are noticed, the investigators should obtain mechanical vapor detectors, such as detector tubes ("Draeger tubes") or detection meters.

14. Name 3 Personal Hygiene Practices that should be followed in the laboratory.

- Wash promptly if skin contact is made with any chemical, regardless of corrosivity.**
- Wear appropriate eye protection at all times.**
- Avoid inhalation of chemicals, do not "sniff" test chemicals**

15. Describe how individual chemicals of less than 15 lbs. (for example a bottle of acetone) are safely transported.

They can be transported via a multiple chemical carrier that has isolated compartments for each bottle or container.

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16. **List** the Emergency Procedure to be followed for a large chemical spill or toxic gas release in the laboratory.

-Evacuate the immediate area.

-Dial "0" advise nature of emergency, and location of emergency

-The operator will notify all employees to evacuate, and will activate the Emergency Response Team.

17. What is the difference between a flammable chemical and a combustible chemical in terms of OSHA and NFPA guidelines.

"Flammable" is generally used to refer to chemicals with a flash point below 100°F. Chemicals with flash points between 100 and 200°F are termed "Combustible".

18. List the Emergency Procedure for an individual chemical exposure in the laboratory.

Dial "0" (Facility Operator), advise nature of emergency, and location of emergency

(E.g., "Person with severe chemical burns, ABC lab").

Operator calls Fire Department Paramedics and notifies Emergency Response Team.

If there are no apparent dangers to yourself of chemical exposure, the following may be performed until the E.R.T./Paramedics arrive:

If a person has been splashed with a chemical, wash them with plenty of water for at least 15 minutes, removing all contaminated clothing (Eyewashes for eye/face splashes, Emergency Shower for all other).

If a person has been overexposed by inhalation, get victim to fresh air (if it is safe for you to enter area), apply artificial respiration if necessary.

19. What is the definition of a "hazardous chemical"?

Substances that can cause adverse health effects.

20. **All** laboratories that handle chemicals must be equipped with eyewashes and safety showers. What is the recommended flowrate of this equipment and how often should they be checked?

30 gallons per minute for fixed showers, 3 gallons per minute for hand-held washes

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